

GenAI & LLMS Assignment -8 Report

Music Generation using Vanilla RNN, GRU and LSTM

Name: Vanchanagiri Alekhya

Enrollment Number: MML2025002

1.Introduction

In this assignment, I implemented and compared three recurrent neural network models for symbolic music generation: **Vanilla RNN, GRU, and LSTM**. The dataset used was the **Nottingham Music Database**, which contains more than 1000 folk tunes written in **ABC notation**. ABC notation is a text-based music representation in which notes, durations, bar lines, repeat symbols, and other musical elements are written using characters.

The aim of this work was to study how well these three models can learn patterns in music and generate new musical sequences. In particular, I wanted to compare them in terms of:

- how well they preserve musical structure
- how well they learn long-term dependencies
- training time and convergence
- model size and computational cost

To ensure a fair comparison, all three models were trained with the same settings: embedding dimension 128, recurrent units 256, sequence length 64, batch size 128, and 30 epochs.

2.Experimental Setup

The ABC music files were first cleaned by removing headers and keeping only the tune bodies. Then character-level sequences of length 64 were created. Each model was trained to predict the next character from the previous 64 characters.

The three architectures used were:

- **Vanilla RNN**
- **GRU**
- **LSTM**

Each model had the same overall structure:

Embedding → Recurrent Layer → Dense Output with Softmax

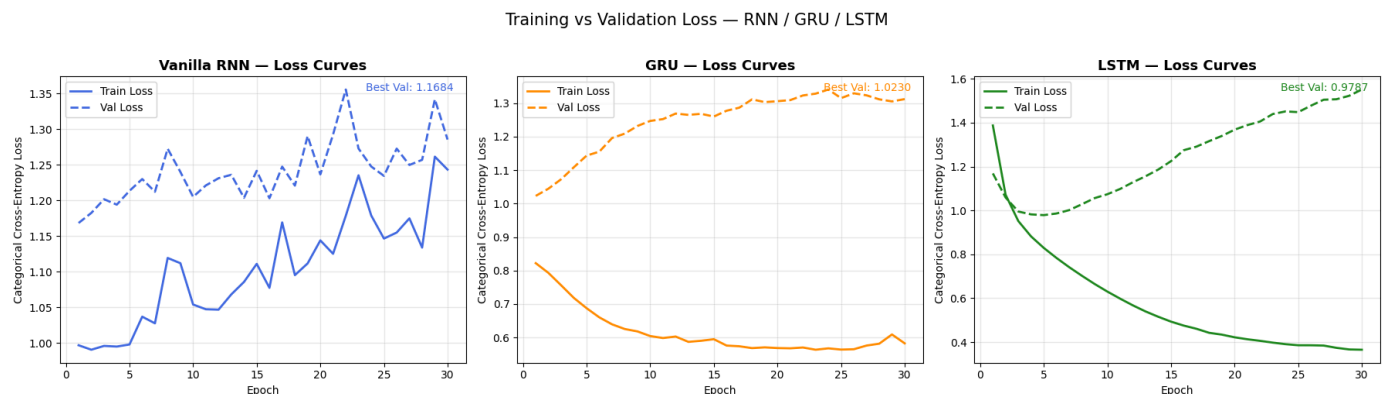
Only the recurrent layer was changed for comparison.

3. Training Results

The training results clearly show differences between the three models.

Model	Best Val Loss	Best Val Accuracy	Training Time (s)	Parameters
Vanilla RNN	1.1684	0.6402	518.2	120,890
GRU	1.0230	0.6863	690.2	318,778
LSTM	0.9787	0.6939	746.8	416,570

From this table, LSTM achieved the **lowest validation loss** and **highest validation accuracy**, which means it learned the music data better than the other two models. GRU came second and performed quite close to LSTM. Vanilla RNN gave the weakest result.



4. Musical Structure Maintenance

Music is not just a random sequence of notes. It contains repeating phrases, rhythmic balance, bar-line structure, and melody flow. So one important question is whether the generated output still looks and feels musical.

4.1 Vanilla RNN

Vanilla RNN gave the weakest musical structure. In many cases, it produced locally valid note patterns, but it struggled to keep the music stable over longer sequences. Its validation loss was also the highest, at **1.1684**, and the validation accuracy was the lowest, at **0.6402**.

For example, at lower temperature like $T = 0.3$, it produced somewhat structured output with repeated note patterns, but as the temperature increased, the output became more unstable. At $T = 1.5$, the generated sequence became much more random and less musically meaningful.

Its structure metrics also show this trend:

- Bar Density at $T = 0.3$: **5.67%**
- Unique Characters at $T = 0.3$: **36.21%**
- At $T = 1.5$, Unique Characters rise to **62.07%**, which suggests much more randomness

So overall, Vanilla RNN could capture some short local patterns, but it could not maintain larger musical coherence.

4.2 GRU

GRU produced much better structure than Vanilla RNN. It was able to keep more regular phrase boundaries and note progressions, especially at moderate temperatures.

Its best validation loss was **1.0230**, and validation accuracy was **0.6863**, which is clearly better than Vanilla RNN. In the generated examples, GRU outputs at $T = 0.3$ and $T = 0.7$ looked more organized and musically meaningful.

Its structure values were also stronger:

- Bar Density at $T = 0.3$: **8.67%**
- Note Ratio at $T = 0.3$: **41.0%**

These values suggest that the generated sequence contained a good amount of note symbols and bar markers, which is a positive sign for musical structure.

At high temperature like $T = 1.5$, GRU also became more random, with Unique Characters increasing to **68.97%**, but at balanced temperatures it performed well.

4.3 LSTM

LSTM gave the best overall musical structure. Among all three models, it generated the most coherent and stable musical sequences. It achieved the **best validation loss of 0.9787** and **best validation accuracy of 0.6939**.

Its generated samples, especially at **T = 0.3** and **T = 0.7**, looked more musically consistent, with clearer patterns and better continuity. Even from the sample output, LSTM appears better at maintaining tune flow and phrase progression.

Its structure analysis also supports this:

- Bar Density at T = 0.3: **9.33%**
- Note Ratio at T = 0.7: **39.0%**
- Note Ratio at T = 1.0: **45.0%**

These values show that LSTM maintains strong musical content and bar structure, especially at low and medium temperatures.

So in terms of structure maintenance, the ranking is clearly:

LSTM > GRU > Vanilla RNN

5. Long-Term Dependency Learning

One of the main challenges in sequence modeling is learning long-term dependencies. In this task, the model must predict the next character based on the previous 64 characters. That means it should remember earlier musical context while generating later notes.

The results show that:

- **LSTM** learned long-term patterns best
- **GRU** also handled long dependencies well
- **Vanilla RNN** struggled the most

This is visible in both the training logs and the plots.

For LSTM, the training accuracy kept improving steadily and finally reached about **0.8770**, while validation accuracy peaked around **0.6939**. This means LSTM learned the training data very well, though it also began to overfit after some epochs.

GRU also learned well, reaching training accuracy above **0.80**, but its validation performance remained slightly below LSTM.

Vanilla RNN, on the other hand, showed unstable behavior. Its validation loss was best in the **first epoch itself**, and then it got worse. That is a strong sign that it could not properly learn deeper musical structure. It mostly learned only very local patterns.

So based on actual results, LSTM was the strongest model for capturing longer musical dependencies.

6. Training Efficiency and Convergence

Training efficiency includes both how fast the model trains and how quickly it reaches a good result.

Training Time

The total training times were:

- Vanilla RNN: **518.2 seconds**
- GRU: **690.2 seconds**
- LSTM: **746.8 seconds**

This shows that Vanilla RNN was the fastest, and LSTM was the slowest.

Convergence behavior

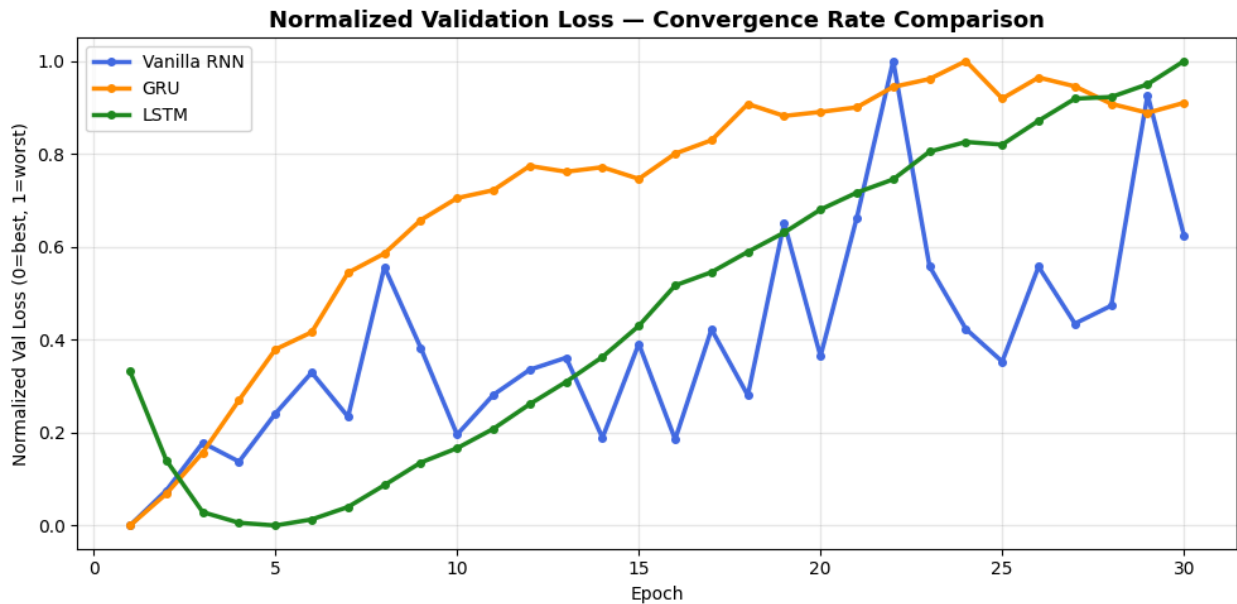
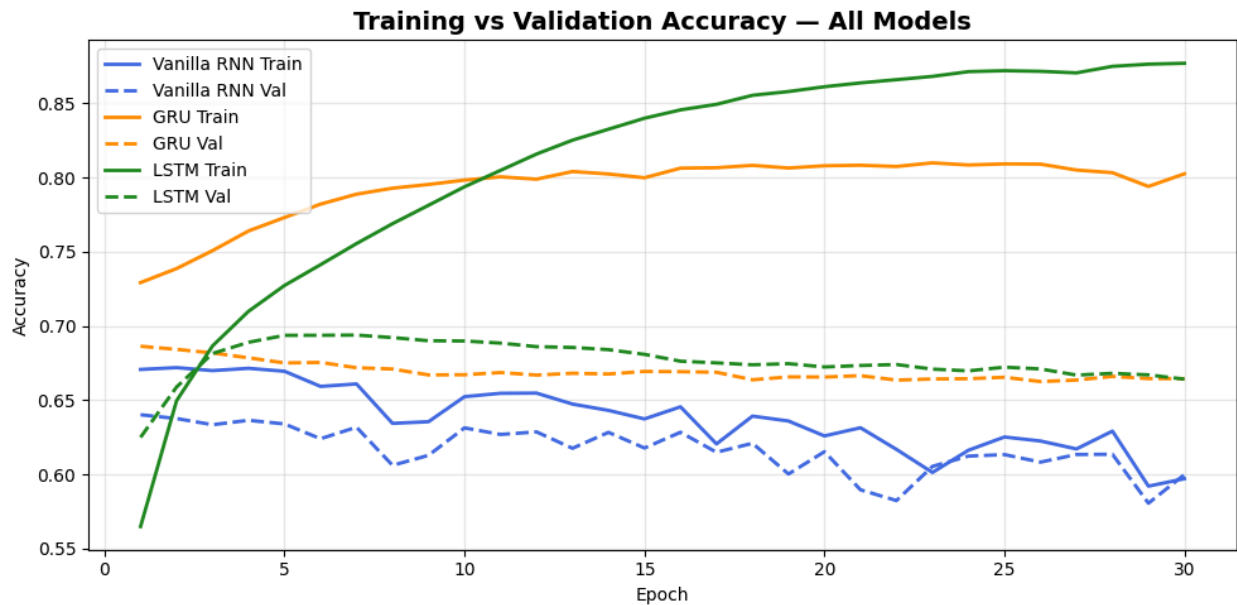
From the loss curves:

- **Vanilla RNN** did not show stable improvement. Validation loss remained high and fluctuated.
- **GRU** learned quickly and consistently, though validation loss slowly worsened after early epochs.
- **LSTM** showed the strongest training improvement and best validation loss in the early stages.

An important observation is that both **GRU and LSTM achieved their best validation loss very early**, and then later started overfitting. In fact:

- GRU best val loss: **1.0230**
- LSTM best val loss: **0.9787**
- Vanilla RNN best val loss: **1.1684**

So although LSTM takes the most time, it gives the best final quality. GRU gives a very good trade-off between speed and performance.



7. Model Size and Computational Requirements

The three models also differ in size:

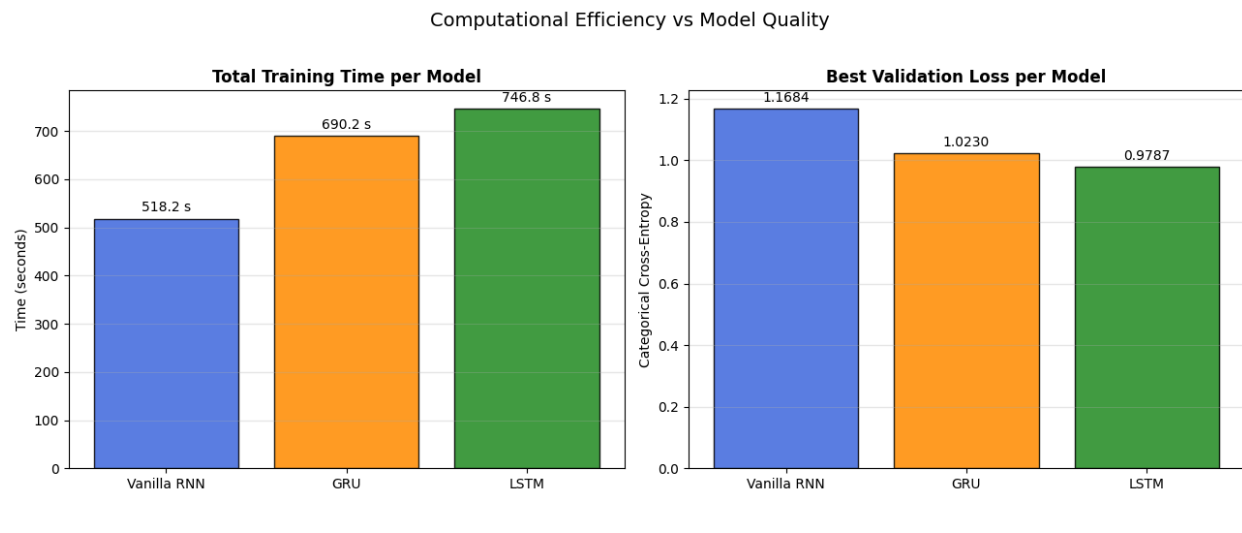
- Vanilla RNN: **120,890 parameters**
- GRU: **318,778 parameters**
- LSTM: **416,570 parameters**

So LSTM is the heaviest model, followed by GRU, then Vanilla RNN.

This means:

- Vanilla RNN is light but weak
- GRU gives a better balance between size and performance
- LSTM gives the best quality but uses the most computation

For practical use, if training speed and parameter count matter a lot, GRU is a very good choice. If the main goal is best music quality, LSTM is better.



8. Temperature Sampling Analysis

Temperature controls the randomness of generated output.

Observations from the generated samples

T = 0.3

This produced the most conservative output. The sequences were more repetitive, but also more structured and safer. For all three models, this temperature gave the most stable musical results.

T = 0.7

This gave the best balance between coherence and variety. The music still looked structured, but it was not too repetitive.

T = 1.0

At this temperature, the output became more creative, but small irregularities started appearing.

T = 1.5

This was the most random setting. All models became noisier, and invalid or strange patterns increased.

Based on your metrics

For example:

- Vanilla RNN Unique Chars at T = 1.5: **62.07%**
- GRU Unique Chars at T = 1.5: **68.97%**
- LSTM Unique Chars at T = 1.5: **53.45%**

This shows that high temperature increases randomness a lot. LSTM still remained somewhat more controlled than GRU at very high temperature.

From the generated examples, **T = 0.7** appears to be the best overall temperature for producing musically pleasing output.

The experimental results match the theory behind these architectures.

Vanilla RNN has no gating mechanism, so it struggles to remember information over long sequences. That is why it performed poorly both in validation metrics and in generated output quality.

GRU and LSTM are better because they include memory control mechanisms. GRU is simpler and more efficient, while LSTM is more expressive and performs slightly better.

From this experiment:

- LSTM gave the best music quality
- GRU gave very strong performance with fewer parameters
- Vanilla RNN mainly serves as a baseline

One limitation of this study is that the music was modeled at the **character level**, not at the note-token level. A token-level representation using notes, durations, and rests might improve generation quality further.

9. Conclusion

This assignment compared Vanilla RNN, GRU, and LSTM for symbolic music generation using ABC notation from the Nottingham dataset.

The main findings are:

- **LSTM performed best overall**, with the lowest validation loss (**0.9787**) and highest validation accuracy (**0.6939**)
- **GRU performed very close to LSTM**, while using fewer parameters and less time
- **Vanilla RNN performed the worst**, especially for maintaining long-range musical structure
- **T = 0.7** gave the best balance between coherence and creativity in generation

So the final ranking is:

LSTM > GRU >> Vanilla RNN

If the goal is best generation quality, LSTM is the best choice. If a balance between efficiency and performance is needed, GRU is the better option.